EXPERIMENT (6)

AIM

To retrieve records from Hostel Management System tables using SELECT with BETWEEN, IN and Aggregate Functions.

DESCRIPTION

BETWEEN selects values within a range. IN checks values from a list. Aggregate functions such as AVG, COUNT, SUM, MIN and MAX perform calculations on groups of rows.

SYNTAX

SELECT column_name FROM table_name WHERE column_name BETWEEN value1 AND value2;

SELECT column_name FROM table_name WHERE column_name IN(value1,value2,...);

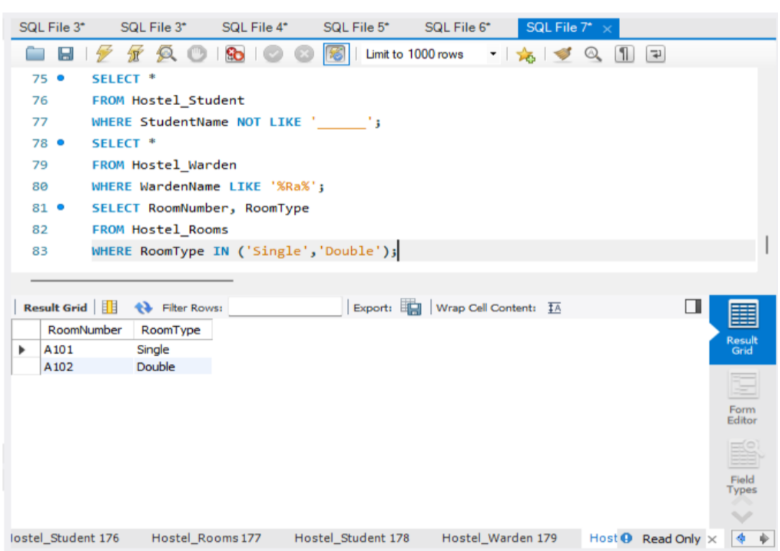
SELECT AVG(column), MIN(column), MAX(column), SUM(column), COUNT(column) FROM table_name;

Questions and Answers

IN & BETWEEN

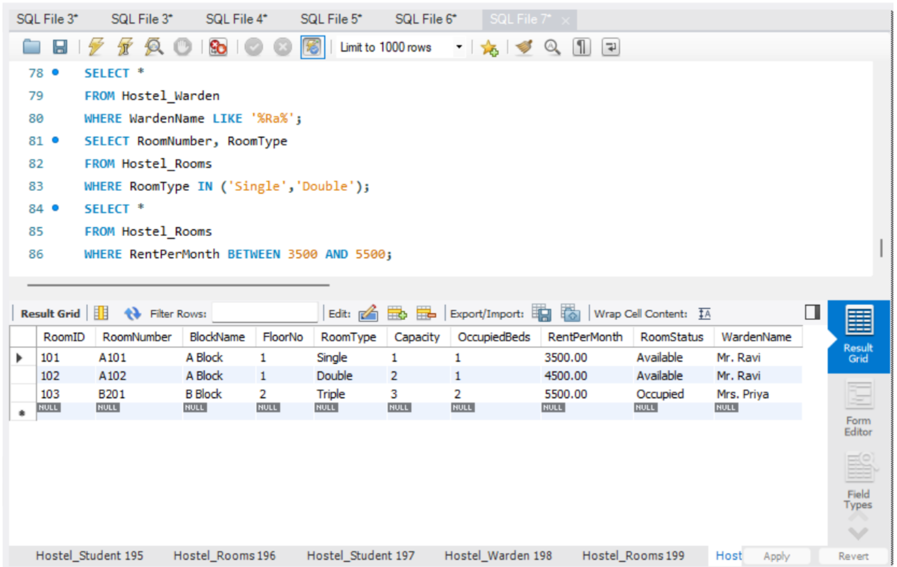
1. List the room numbers of rooms whose type is 'Single' or 'Double'.

SELECT RoomNumber, RoomType FROM Hostel_Rooms WHERE RoomType IN ('Single','Double');



2. Display rooms whose monthly rent is between 3500 and 5500.

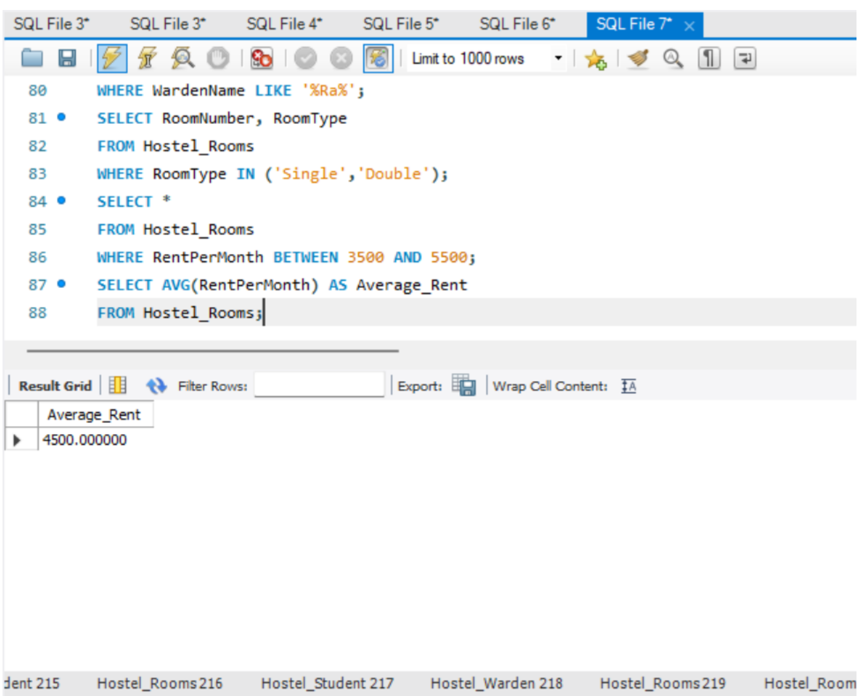
SELECT * FROM Hostel_Rooms WHERE RentPerMonth BETWEEN 3500 AND 5500;



AGGREGATE

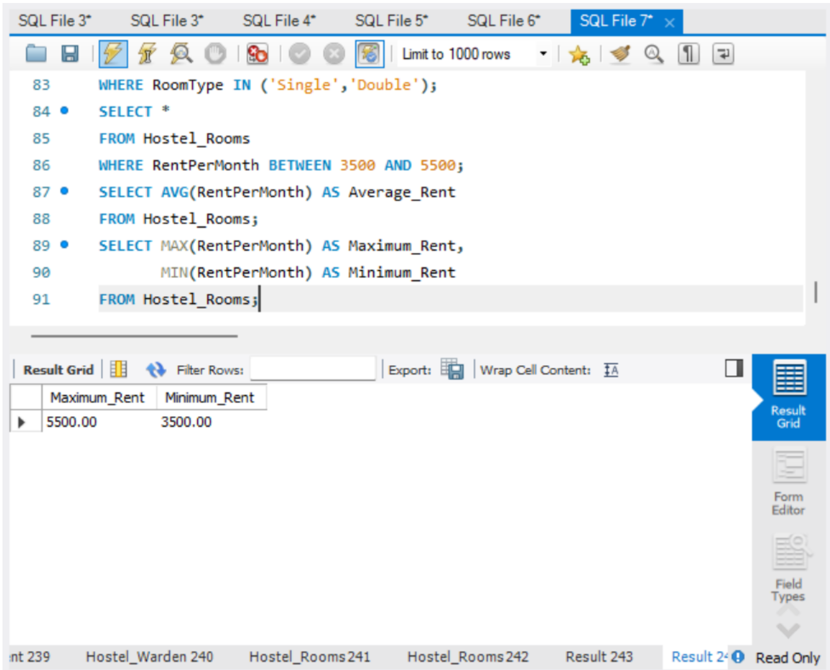
1. Find the average monthly rent of all hostel rooms.

SELECT AVG(RentPerMonth) AS Average_Rent FROM Hostel_Rooms;



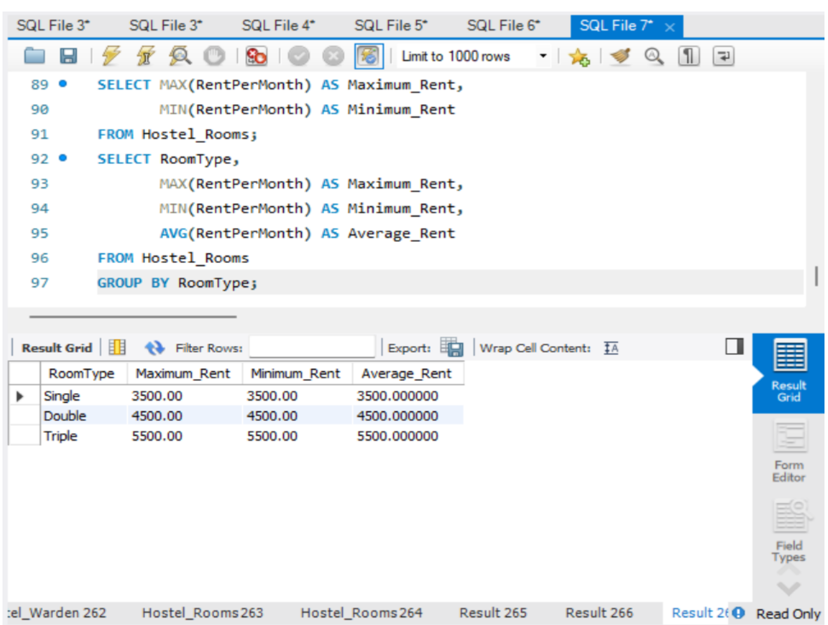
2. Display the maximum and minimum monthly rent.

SELECT MAX(RentPerMonth) AS Maximum_Rent, MIN(RentPerMonth) AS Minimum_Rent FROM Hostel_Rooms;



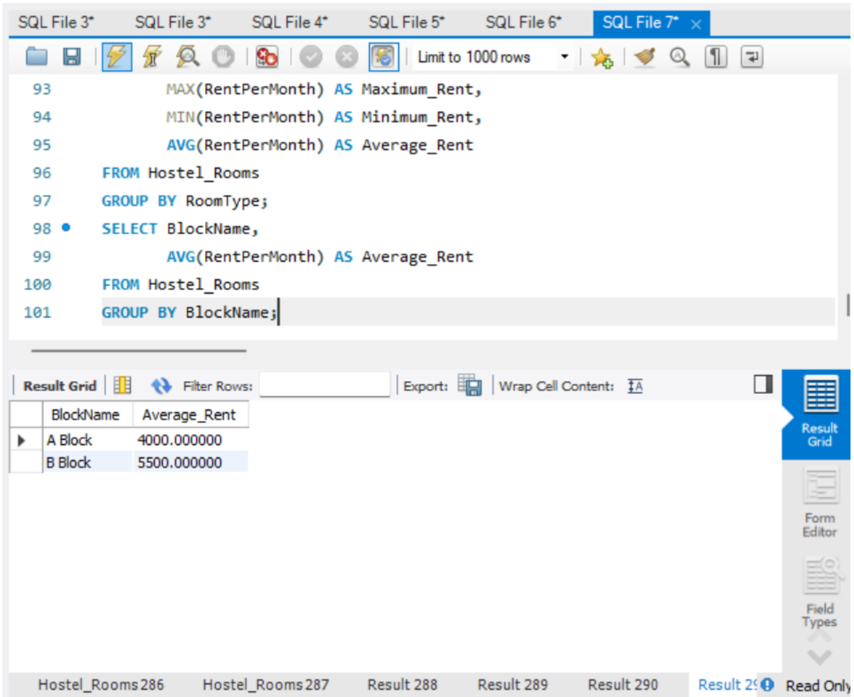
3. Display the maximum, minimum and average rent for each room type.

SELECT RoomType, MAX(RentPerMonth) AS Max_Rent, MIN(RentPerMonth) AS Min_Rent, AVG(RentPerMonth) AS Avg_Rent FROM Hostel_Rooms GROUP BY RoomType;



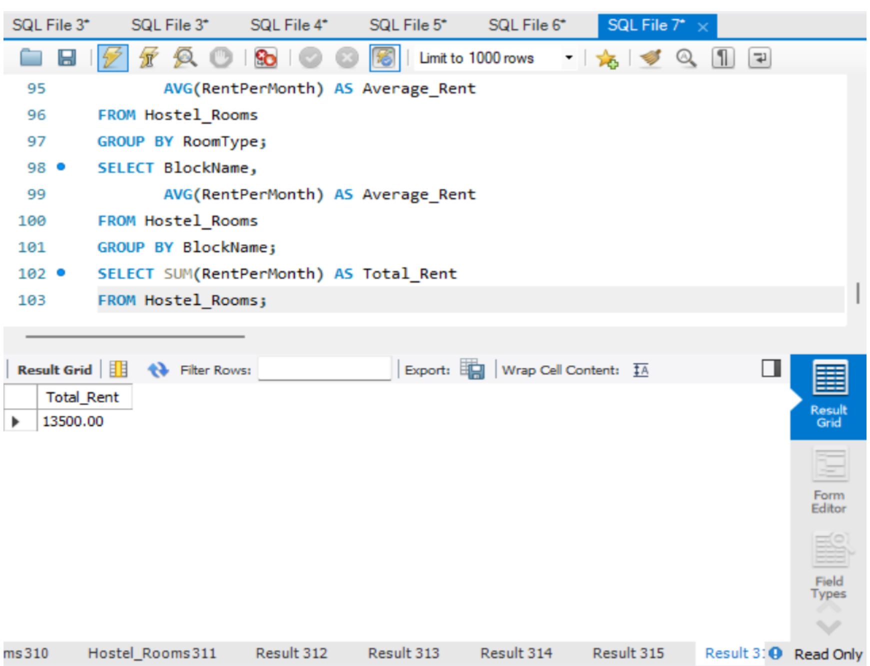
4. Display block name and average monthly rent.

SELECT BlockName, AVG(RentPerMonth) AS Avg_Rent FROM Hostel_Rooms GROUP BY BlockName;



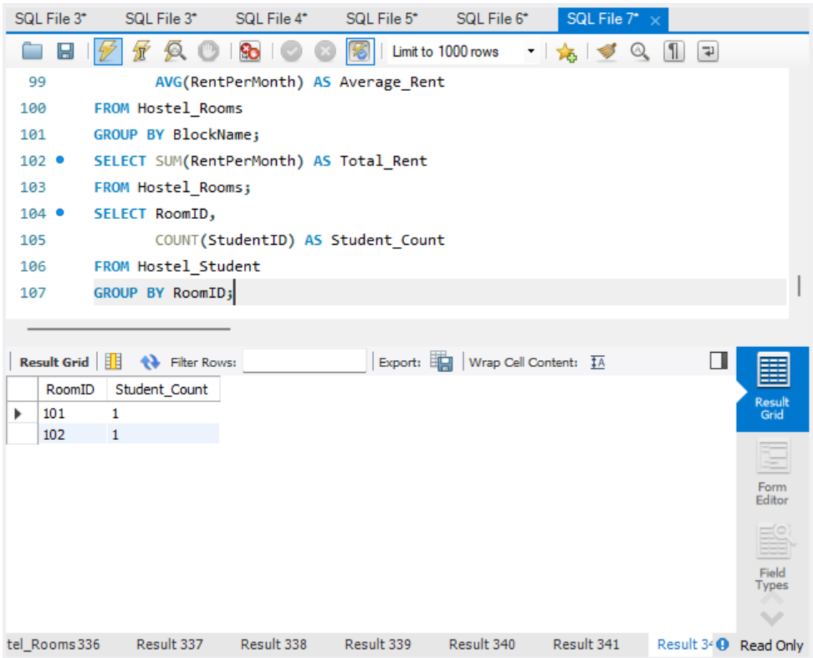
5. Calculate the total monthly rent of all rooms.

SELECT SUM(RentPerMonth) AS Total_Rent FROM Hostel_Rooms;



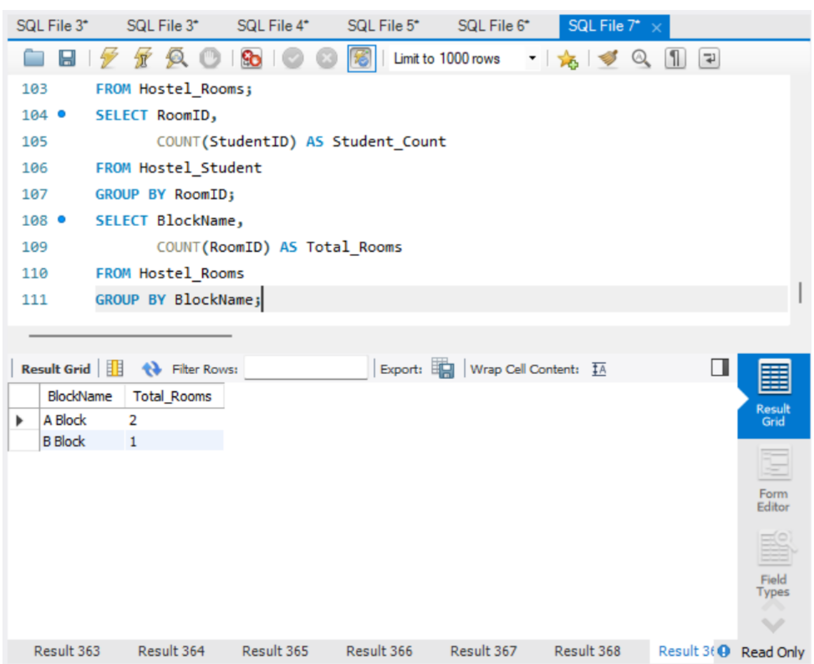
6. Display the number of students staying in each room.

SELECT RoomID, COUNT(StudentID) AS Student_Count FROM Hostel_Student GROUP BY RoomID;



7. Display the number of rooms available in each block.

SELECT BlockName, COUNT(RoomID) AS Total_Rooms FROM Hostel_Rooms GROUP BY BlockName;



RESULT

Thus, the records were successfully retrieved using SELECT statements with BETWEEN, IN and Aggregate Functions on the Hostel Management System tables.